

AI Classification Model - Image

1. Creación inicial del modelo de clasificación

Configuración inicial del proyecto en SenseCraft AI para registrar las primeras clases de objetos detectables.

SenseCraft

SenseCraft AI

Home

Applications

Models

Community

English

R rauljauregui

Image Classification Detection

Audio Classification Detection

Image Object Detection

Generate AI Classification Model - Image

Step 1: Collect Classification Data

Sillas

...

Please collect image samples

Class 2

...

Please collect image samples

Add a Class

XIAO ESP32S3 Sense

Disconnect

2. Registro de muestra “Sillas”

Recolección de muestras fotográficas correspondientes a la categoría “Sillas” para entrenar el modelo.

SenseCraft

SenseCraft AI

Home

Applications

Models

Community

English

R rauljauregui

Step 1: Collect Classification Data

Sillas

...

Please collect image samples

Class 2

...

Please collect image samples

Add a Class

XIAO ESP32S3 Sense

Disconnect

Hold to Record

Stop

3. Registro de muestra “Computadoras”

Recolección de muestras fotográficas correspondientes a la categoría “Computadoras” para entrenar el modelo.

SenseCraft | SenseCraft AI

Home Applications **Models** Community

English rauljauregui

67 Image Samples

Computadoras Laboratorios

83 Image Samples

Import Dataset Export Dataset Clear

Step 2: Training

4. Registro de muestra “Impresoras 3D”

Recolección de muestras fotográficas correspondientes a la categoría “Impresoras 3D” para entrenar el modelo.

SenseCraft | SenseCraft AI

Home Applications **Models** Community

English rauljauregui

Step 1: Collect Classification Data

Computadoras Laboratorios

125 Image Samples

Impresoras 3D

166 Image Samples

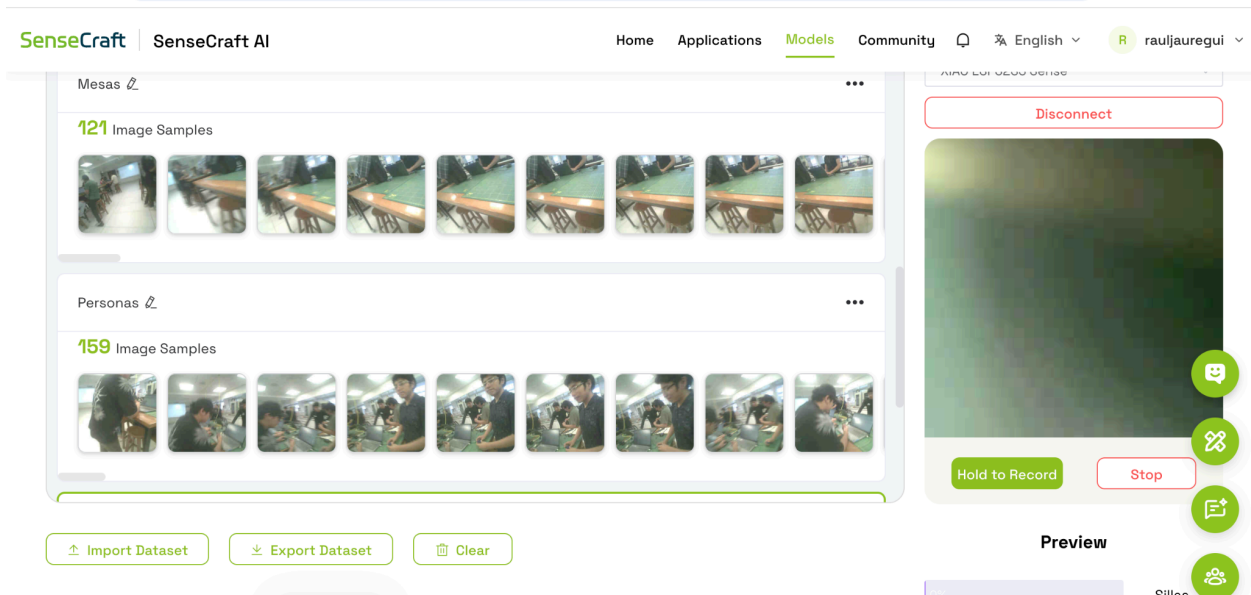
XIAO ESP32S3 Sense

Disconnect

Hold to Record Stop

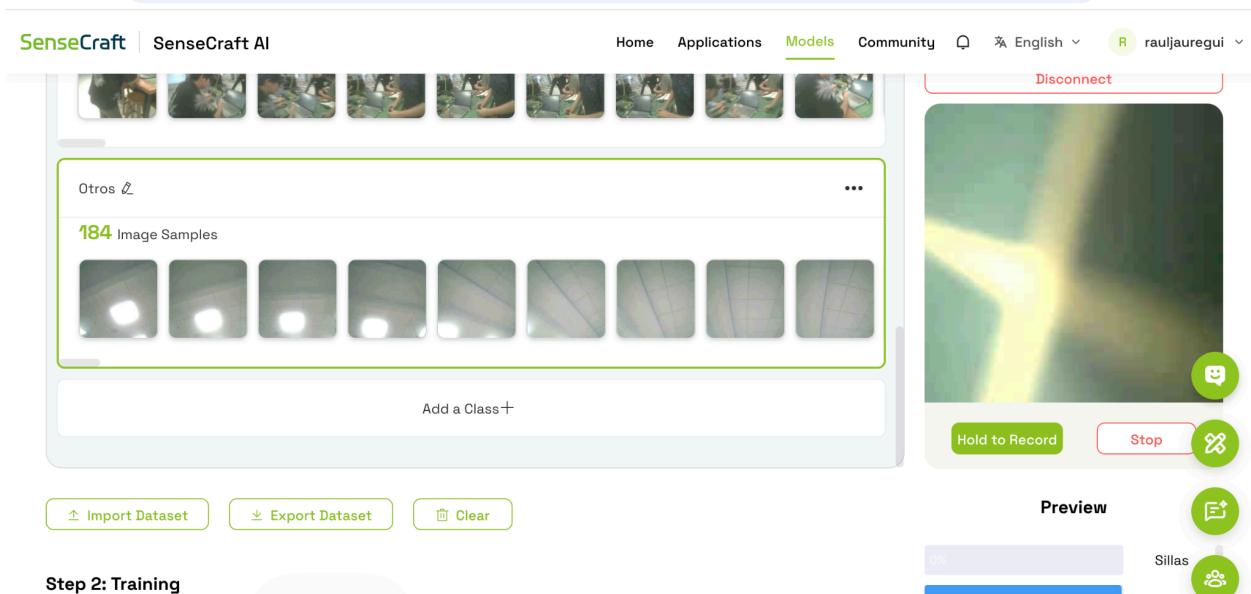
5. Registro de muestra “Mesas” y “Personas”

Recolección de muestras fotográficas correspondientes a la categoría “Mesas” y “Personas” para entrenar el modelo.



6. Registro de muestra “Otros”

Recolección de muestras fotográficas correspondientes a la categoría “Otros” para entrenar el modelo.



7. Entrenamiento preliminar del modelo

Inicio del proceso de entrenamiento utilizando las imágenes clasificadas previamente en cada categoría

The screenshot shows the SenseCraft AI web interface. At the top, there's a navigation bar with 'SenseCraft AI' and links for 'Home', 'Applications', 'Models' (highlighted), 'Community', and a language dropdown set to 'English'. A user profile 'rauljauregui' is visible. Below the navigation bar, there are buttons for 'Hold to Record' and 'Stop'. The main content area has three buttons: 'Import Dataset', 'Export Dataset', and 'Clear'. The 'Step 2: Training' section shows a dropdown menu with 'XIAO ESP32S3 Sense' and a green button labeled 'Training in progress'. Below this, there's an 'Advanced' dropdown. The 'Step 3: Deploy and Preview' section has a message: 'Please train a new model or select a model from history to deploy'. The 'History Models(0)' section is empty. On the right side, there's a vertical sidebar with four green circular icons: a chat bubble, a gear, a document with a plus sign, and a user icon. The word 'Expand' is partially visible next to the third icon.

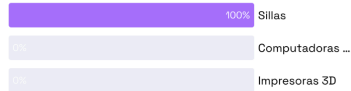
8. Validación de reconocimiento en tiempo real

Evaluación del modelo mediante pruebas en vivo para verificar la detección correcta de objetos.

The screenshot shows the SenseCraft AI web interface during a real-time evaluation. The top navigation bar is the same as in the previous screenshot. The main content area is divided into three sections. On the left, there's a 'Serial Port' section with an 'Audio' label. In the center, there's a video feed showing a hand holding a small device over a laptop. Below the video feed, there's a list of detected objects with their confidence scores: 'Sillas' (chairs) with a low score, 'Computadoras ...' (computers) with a 98% score, and 'Impresoras 3D' (3D printers) with a low score. On the right, there's a 'WIFI & MQTT' section with a 'Configuration' button and two status indicators: 'IP Address: Loading' and 'Service Status: Loading'. Below this is a 'Device Logger' section showing a list of performance metrics in JSON format, such as 'perf:{"preprocess":107,"inference":431,"postprocess":0}'. On the far right, there's a vertical sidebar with four green circular icons: a chat bubble, a gear, a document with a plus sign, and a user icon.

Serial Port

Audio



WIFI & MQTT

Configuration

IP Address: Loading

Service Status: Loading

Device Logger

```
perf:{"preprocess":109,"inference":433,"postpr
ocess":0}
perf:{"preprocess":103,"inference":434,"postpr
ocess":0}
perf:{"preprocess":103,"inference":428,"postpr
ocess":0}
perf:{"preprocess":110,"inference":428,"postpr
ocess":0}
perf:{"preprocess":109,"inference":426,"postpr
ocess":0}
perf:{"preprocess":110,"inference":431,"postpr
ocess":0}
```

Serial Port

Audio



WIFI & MQTT

Configuration

IP Address: Loading

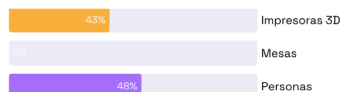
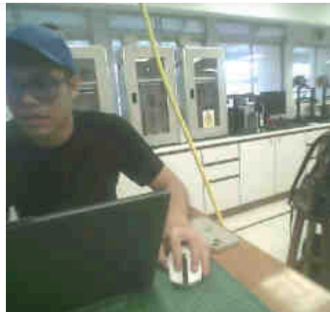
Service Status: Loading

Device Logger

```
perf:{"preprocess":108,"inference":432,"postpr
ocess":0}
perf:{"preprocess":102,"inference":435,"postpr
ocess":0}
perf:{"preprocess":100,"inference":433,"postpr
ocess":0}
perf:{"preprocess":102,"inference":428,"postpr
ocess":0}
perf:{"preprocess":108,"inference":428,"postpr
ocess":0}
perf:{"preprocess":108,"inference":426,"postpr
ocess":0}
```

Serial Port

Audio



WIFI & MQTT

Configuration

IP Address: Loading

Service Status: Loading

Device Logger

```
perf:{"preprocess":77,"inference":454,"postpro
cess":0}
perf:{"preprocess":82,"inference":451,"postpro
cess":0}
perf:{"preprocess":85,"inference":452,"postpro
cess":0}
perf:{"preprocess":84,"inference":449,"postpro
cess":0}
perf:{"preprocess":83,"inference":455,"postpro
cess":1}
perf:{"preprocess":92,"inference":457,"postpro
cess":0}
```

9. Monitoreo mediante MQTT

Configuración de transmisión de datos y visualización de inferencias utilizando el protocolo MQTT.

The screenshot shows the SenseCraft AI web interface. The top navigation bar includes 'SenseCraft AI', 'Home', 'Applications', 'Models' (highlighted), 'Community', a search icon, 'English', and a user profile 'rauljauregui'. On the left sidebar, under the 'Output' section, 'MQTT' is selected. The main content area is titled 'Output data through MQTT' and includes a description: 'SenseCraft MQTT, which helps you quickly check whether the inference data of the device model is sent successfully'. It lists three steps: 'Step 1: Configure MQTT Service' (with a 'Configuration' button), 'Step 2: Subscribe to the topic' (showing a subscription to 'sscma/v0/Clasificacion Equipo 2/tx' with a checkmark), and 'Step 3: Send the wake-up command and receive messages'. A 'Disconnect' button is visible in the top left of the main area. On the right side, there are four green circular icons: a chat bubble, a gear, a document with a plus sign, and a group of people.

10. Configuración de conexión del dispositivo

Ajuste de parámetros de red, host y credenciales necesarias para la comunicación del sistema inteligente.

The screenshot shows the 'Configuration' dialog box in the SenseCraft AI interface. The dialog has a title bar with a close button. It contains the following fields and controls: 'SSID' (text input with 'Nico \$'), 'Password' (password input with a toggle for visibility), 'Encryption' (dropdown menu set to 'AUTO'), 'MQTT' (toggle switch set to 'Yes' with a link to 'MQTT Communication Workflow'), 'Mode' (dropdown menu set to 'Custom Mode'), 'Host' (text input with 'rp10.rcr-labs.com'), 'Port' (text input with '1883'), 'ClientId' (text input with 'e2'), and 'Screenshot' (text input with 'Alumno'). The background shows the same interface as the previous screenshot, but the 'MQTT' option in the sidebar is highlighted.

Output

MQTT

GPIO

Serial Port

Audio

* Host wss:// rp10.rcr-labs.com

* Port 1883

Path /mqtt

ClientId user-1778606299462

User Name Alumno

Password

Cancel

Apply

Screenshot

Send

